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6.3.2 Backward error analysis of dot product: general case

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Let's move on to the general case of dot product, where we take the dot product of

two vectors of size n each. And if you write it out, then the exact result would

require the given multiplications and additions. And we're going to assume that

Advo

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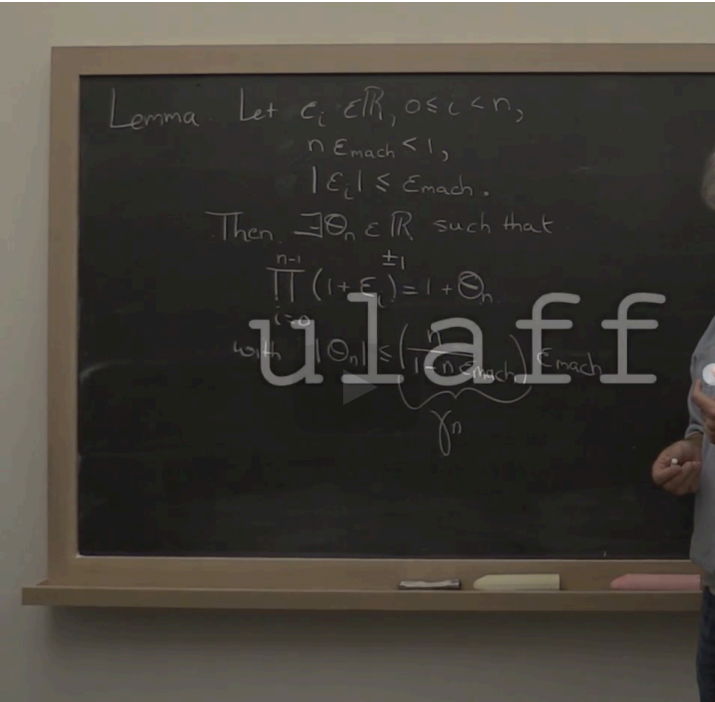
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“

1 plus theta_n
is equal to the
product of all of
those factors. Okay?
Moreover you can
prove
that if each of the
epsilons in magnitude
is less than machine
epsilon,
and notice that the
absolute value of
theta n is then less
than something
times the machine
epsilon where that
something
is a function of n. The
proof of this --well,
you can kind of look
at that and say,
"Well probably I need
to prove that by

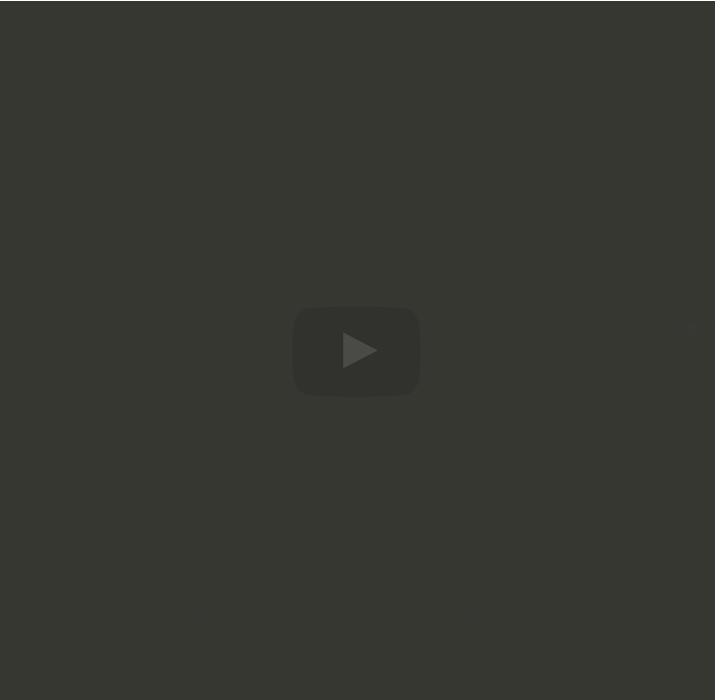
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“

With this, we can take
this really ugly
expression if we had
previously and
write it much more
concisely as the
computed value
when you do your dot
products with vectors
of size n is given by
the exact dot
product, except that
now
we throw these
factors in of 1 plus
theta_n for the first
term, 1 plus
theta_n for the

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Homework 6.3.2.5

1/1 point (graded)
Prove Lemma 6.3.2.5.

☒ done



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